

CHRISTCHURCH, New Zealand

WMO: 937810

Lat: 43.4897S

Lon: 172.5269E

Elev: 117

StdP: 14.63

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.9	28.7	24.5	18.7	34.8	26.3	20.3	35.7	25.4	48.1	22.7	48.3	2.5	290	0.613

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.3	82.7	62.5	79.0	61.0	75.4	60.0	65.3	76.2	63.8	73.6	62.4	71.0	14.7	300

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
62.3	84.2	68.1	60.7	79.6	66.1	59.2	75.3	64.7	30.3	76.4	29.2	73.9	28.2	71.4	72.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
25.2	22.4	20.0		23.4	90.4	1.5	2.7	22.3	92.3	21.4	93.9	20.6	95.4	19.5	97.3	
			DB	22.9	69.0	1.4	1.6	21.9	70.2	21.0	71.1	20.3	72.0	19.2	73.2	
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	61.8	61.6	58.6	53.3	49.1	43.9	43.0	45.1	48.7	51.9	55.4	60.0
	DBStd	8.47	6.07	5.94	5.58	5.23	5.27	4.90	4.61	4.62	5.07	5.17	5.94	5.72
	HDD50	830	2	3	3	22	81	196	223	164	83	38	14	1
	HDD65	4619	137	125	211	353	494	634	681	616	489	408	293	178
	CDD50	1795	369	328	271	120	52	11	7	13	44	95	175	310
	CDD65	109	38	30	13	1	0	0	0	0	0	5	22	
	CDH74	1087	344	274	141	18	2	0	0	0	10	72	226	
	CDH80	252	93	78	26	0	0	0	0	0	0	10	45	
Wind	WSAvg	8.9	10.7	10.2	9.3	7.9	7.4	6.8	6.9	7.7	8.8	9.8	10.3	10.8
Precipitation	PrecAvg	24.1	1.7	1.5	2.0	2.0	2.4	2.1	2.6	2.5	1.7	1.8	1.9	1.8
	PrecMax	35.9	5.5	4.5	6.8	7.8	6.5	6.6	7.1	6.7	4.4	5.4	4.6	6.1
	PrecMin	12.1	0.3	0.2	0.1	0.3	0.5	0.2	0.0	0.5	0.1	0.1	0.4	0.0
	PrecStd	5.2	1.2	0.9	1.6	1.7	1.3	1.4	1.9	1.6	1.2	1.1	1.0	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.6	87.8	83.9	76.9	71.5	64.5	64.0	65.9	71.4	75.3	80.8	84.6
		MCWB	63.6	64.4	62.6	58.2	57.0	52.6	52.1	52.4	55.3	58.5	60.9	62.4
	2%	DB	82.4	82.1	78.4	71.5	66.6	60.4	59.0	62.2	66.5	71.2	75.3	80.3
		MCWB	62.2	63.2	61.2	57.3	54.5	50.6	49.3	50.7	53.3	56.3	58.4	61.1
	5%	DB	77.3	76.7	73.3	66.6	62.8	56.8	55.3	58.6	62.9	66.6	71.6	75.5
		MCWB	60.8	61.7	59.4	55.6	53.1	49.3	47.4	49.0	51.8	54.4	57.3	59.8
	10%	DB	73.2	71.8	69.0	62.9	58.9	53.4	51.9	54.9	59.3	62.9	67.6	71.5
		MCWB	60.4	60.8	58.6	54.9	51.7	48.0	46.6	47.6	50.0	52.8	55.8	58.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.6	67.4	65.7	60.8	58.7	54.1	52.7	53.5	57.1	60.2	63.0	66.4
		MCDB	78.8	79.1	76.8	69.6	68.0	61.1	61.2	63.0	67.6	71.2	74.9	78.7
	2%	WB	65.2	65.2	63.3	58.9	56.3	51.9	50.3	51.3	54.7	57.4	60.6	63.6
		MCDB	75.5	75.9	72.6	66.8	63.2	57.5	55.5	58.7	64.3	67.9	71.6	74.5
	5%	WB	63.6	63.5	61.3	57.4	54.6	50.2	48.9	49.9	52.7	55.5	58.6	61.8
		MCDB	72.8	72.5	68.7	64.3	60.5	54.2	53.2	55.7	61.1	64.6	68.5	71.2
	10%	WB	62.0	62.0	59.9	55.8	52.9	48.8	47.3	48.6	51.1	53.8	56.8	60.1
		MCDB	69.8	69.3	66.3	61.7	58.2	52.3	51.1	53.1	57.6	61.7	65.5	68.2
Mean Daily Temperature Range	5% DB	MDBR	18.3	18.0	18.5	18.1	17.7	17.3	17.1	17.5	18.8	18.9	19.5	18.2
		MCDBR	27.0	26.6	26.6	25.4	24.0	22.8	22.9	24.8	25.7	26.3	26.5	26.3
		MCWBR	11.2	11.0	12.2	13.8	14.3	15.0	15.0	15.2	14.8	14.4	13.0	11.5
	5% WB	MCDBR	20.7	21.4	20.6	20.7	20.0	18.6	18.9	20.0	22.2	22.9	23.5	21.2
MCWBR		9.7	9.9	10.6	12.2	12.8	13.0	13.1	13.2	13.7	13.5	12.5	10.4	
Clear-Sky Solar Irradiance	taub	0.334	0.327	0.311	0.301	0.284	0.276	0.275	0.289	0.305	0.317	0.319	0.328	
	taud	2.527	2.566	2.602	2.611	2.629	2.611	2.626	2.604	2.529	2.520	2.538	2.553	
	Ebn at Noon	314	308	298	279	262	252	263	279	295	308	317	318	
	Edh at Noon	34	31	27	23	19	17	18	23	29	32	34	34	
All-Sky Solar Radiation	RadAvg	1997	1788	1370	891	583	447	532	770	1165	1606	1957	2028	
	RadStd	170	169	61	89	32	23	38	48	70	110	130	155	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.59	+0.43	+0.74	+1.33	N/A	+0.45	-72	N/A	N/A	N/A
Regional (3 neighbors)		N/A	N/A	+0.56	N/A	N/A	+0.54	-70	N/A	N/A	N/A

Nomenclature: See separate page